

StockSense Pro

AI-based retail shelf intelligence

DRAFT TEMPLATE — student details and live links still required.

Submission field	Details
Student's full name	[TO BE COMPLETED]
Candidate registration number	[TO BE COMPLETED]
School name	[TO BE COMPLETED]
CRS name	Artificial Intelligence
Course name	Machine Learning and Deep Learning
GitHub repository	[TO BE COMPLETED]
Live Streamlit application	[TO BE COMPLETED]

Problem and solution

Manual shelf checks can miss low stock. This project detects cereal, coffee, juice, milk and water in an uploaded photograph, counts visible items, and converts counts into threshold-based stock alerts. It identifies broad categories, not brands or complete store inventory.

Implementation

A balanced 500-image subset of the Freiburg Groceries detection extension was split into 350 training, 75 validation and 75 test images. Each category has 100 images. Source Pascal VOC boxes were converted to YOLO format; exact pixel duplicates were excluded before splitting. Augmentations affect only training data.

YOLO11n was initialized from COCO weights and fine-tuned for 30 epochs at 224 x 224, batch 16, AdamW learning rate 0.001, seed 42, with the first 10 layers frozen. Best validation weights were exported to ONNX for lightweight CPU deployment. The Streamlit app includes annotations, stock metrics, priority alerts, exports and session history.

Measured evaluation

Metric — 75 held-out test images	Result
App precision at confidence 0.25	68.75%
App recall at confidence 0.25	90.06%
F1	77.97%
Count MAE / category / image	0.2053
Exact five-category count rate	48.00%
Ultralytics test mAP@0.50	86.58%
Ultralytics test mAP@0.50:0.95	64.96%

App detection metrics use a confidence threshold of 0.25, class-aware NMS at 0.45, and one-to-one matches at IoU ≥ 0.50 . Standard Ultralytics mAP is a separate evaluator. The test set was not used for choosing the best epoch. Full metrics and per-image counts are included in reports/.

Stock decisions

0 detected items: out of stock; 1-3: low stock; 4 or more: in stock. This resolves the overlapping count-3 ranges in the assignment. Users select expected categories and adjust thresholds/targets. Zero detections prompt a shelf check rather than proving absence.

Testing and limitations

Tests cover threshold boundaries, missing categories, invalid images, normalization, suppression, model inference, split integrity and Streamlit controls. A low-light test reduced exact count agreement from 48% to 40%; the cluttered subset reached 36.36%. Small or obscured products, similar packaging and incomplete background annotations remain limitations. The model is an educational prototype; its metrics do not establish accuracy on new stores. Human feedback and live-cloud testing must be recorded after deployment.

References and assistance

Jund et al. (2016), The Freiburg Groceries Dataset. <https://arxiv.org/abs/1611.05799>

Detection annotation extension: <https://github.com/aleksandar-aleksandrov/groceries-object-detection-dataset>

Redmon et al. (2016), You Only Look Once. <https://arxiv.org/abs/1506.02640>

Ultralytics YOLO11 documentation: <https://docs.ultralytics.com/models/yolo11/>

Developed with Codex assistance. Review the code and disclose assistance according to school requirements. Source dataset and model credits are preserved in the repository.

Application screenshot

StockSense PRO

RETAIL OPERATIONS
WORKSPACE

Shelf settings

Choose the categories this shelf is expected to carry.

Expected products

Cereal

Coffee

Juice

Milk

Low stock: at most

3

Target visible units per category

6

Detection settings

Five trained categories

Cereal · Coffee · Juice · Milk · Water

Counts describe visible products in one image. Occluded products and backroom stock are not counted.

Educational prototype · YOLO11n

Deploy

Shelf intelligence

Review visible stock, inspect detections, and plan the next refill.

Shelf scan

Session history

Model & evidence

Image workspace

Example image

Upload my image

Example

Milk shelf

Real shelf photograph from the evaluation dataset. Attribution is included in Model & evidence.

Analyze shelf

Visible products

5

Out of stock

4

Low stock

0

In stock

1

5 expected categories · 0.08s inference · counts are image estimates

Visual shelf check

Detected products

Original image

4 Milk 75%

5 Milk 60%

3 Milk 83%

1 Milk 92%

2 Milk 84%

Boxes show detected items. Their colors match the category's stock status. Empty categories appear in the alert list.

Category overview

Product	Visible	Status	Top-up to target
cereal	0	Out of stock	6
coffee	0	Out of stock	6

cereal

coffee

juice

milk

Stock rules

0 Out of stock

1-3 Low stock

4+ In stock

A zero is a restocking check, not proof of absence. Confirm the image covers the shelf and the model hasn't missed an item.

Restocking priorities

Cereal

0 VISIBLE · OUT OF STOCK

Check shelf; no items detected · Top-up to target: 6

Coffee

0 VISIBLE · OUT OF STOCK

Check shelf; no items detected · Top-up to target: 6

Juice

0 VISIBLE · OUT OF STOCK

Check shelf; no items detected · Top-up to target: 6

Water

0 VISIBLE · OUT OF STOCK

Check shelf; no items detected · Top-up to target: 6

Milk

5 VISIBLE · IN STOCK

Stock sufficient · Top-up to target: 1

Top-up quantities below use your chosen target. Sales performance cannot be inferred from a single shelf image.

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